BASICS OF COMPUTER NETWORKS

DAY 02

Cabling, Standards & Reference Models

Foundations — what the network is made of, and the rules everyone agreed on

From the Industry to the Classroom

Building the IT's Future, Today and Together!

LECTURE 1 · HOUR 1

Cabling & Speed

Three cable families, the Cat numbering, and why throughput is not the same as bandwidth

Today's Roadmap

What you'll walk out knowing

01

Cable Families

Copper, fiber, and the physics in between

02

Cat Numbering

Twisted-pair categories from 5e to 8

03

Speed Vocabulary

Bandwidth vs throughput vs latency

04

GNS3 + VirtualBox

Use both like a pro

05

OSI vs TCP/IP

The standards debate, settled

★ **KEY TAKEAWAY**

Yesterday: what a network IS. Today: what it's MADE OF — and what RULES everyone agreed to follow.

Cable Families (1/2)

Three families, one job: carry information as a physical change

COPPER

Twisted-Pair

- Voltage wiggles in pairs
- RJ-45 plug
- UTP (Unshielded) — cheap default
- STP (Shielded) — high-EMI sites
- What runs in 99% of offices

COPPER

Coaxial

- Center conductor + braided shield
- Mostly retired from LANs
- Still on cable-Internet drops
- Used by ISPs (DOCSIS)
- Useful to recognize, not to deploy

GLASS

Fiber-Optic

- Light pulses in glass
- SMF (Single-Mode) — laser, km, telecom
- MMF (Multi-Mode) — LED/VCSEL, 100s m
- Immune to EMI
- Standard inside data centers

★ **KEY TAKEAWAY**

Pick the cable that matches the ENVIRONMENT, not the budget. Fiber wins anywhere EMI or distance is a problem.

Cable Families (2/2)

The differential-signaling trick — why UTP works at all

THE PROBLEM

A copper wire next to a fluorescent light or motor picks up EMI noise — random voltages riding on top of your signal.

THE TRICK

Send the SAME signal on two wires — one normal, one inverted. Twist them tightly together.

EMI hits both wires equally → at the receiver, subtract one from the other and the noise CANCELS while the signal DOUBLES.

WHY IT MATTERS

More twists per inch = better noise cancellation = higher Cat number.

INSIDE A TWISTED PAIR

```
Sender                          Receiver
  +V ──╲ ╱── ╲ ╱── ╲ ╱──  +V
        X     X     X
  -V ──╱ ╲── ╱ ╲── ╱ ╲──  -V

  EMI noise hits BOTH wires
  Receiver subtracts: noise → 0
  signal → doubled

More twists = more noise rejection
```

★ KEY TAKEAWAY

The twist is the technology. Cat 5e and Cat 6 differ mostly in HOW MANY TIMES per inch the wires twist together.

Twisted-Pair Categories (1/2)

The Cat number is the speed cap of the link

Category	Max Speed	Max Bandwidth	Max Distance	Common Use
Cat 5	100 Mbps	100 MHz	100 m	Legacy — replace it
Cat 5e	**1 Gbps**	**100 MHz**	**100 m**	**Office workhorse**
Cat 6	1 Gbps (10 G ≤ 55 m)	250 MHz	100 m	New offices
Cat 6a	**10 Gbps**	**500 MHz**	**100 m**	**10 G to the desk**
Cat 7	10 Gbps	600 MHz	100 m	Niche, EU shielded
Cat 8	**25–40 Gbps**	**2000 MHz**	**30 m**	**Top-of-rack DC**

★ **KEY TAKEAWAY**

The cable category sets the CEILING for the entire link. A Cat 5 cable strangles a 1 Gbps NIC.

Twisted-Pair Categories (2/2)

The 100-meter rule and the T568A/B pinouts

THE 100-METER RULE

Almost every twisted-pair category caps at 100 m end-to-end (cable + patch leads).

Beyond 100 m, the signal attenuates and timing drifts. The link still LIGHTS UP — but you'll see CRC errors, retransmits, and silent slowdowns.

Cat 8 stops at 30 m on purpose — it's a top-of-rack cable that REPLACES short fiber runs.

Going farther?

Use fiber. SMF runs kilometers, MMF runs hundreds of meters.

T568A vs T568B PINOUTS

```
T568B  (most common)
  1  Orange/White
  2  Orange
  3  Green/White
  4  Blue
  5  Blue/White
  6  Green
  7  Brown/White
  8  Brown

Same on BOTH ends → straight-through
DIFFERENT on each end → crossover
(rare today – auto MDI-X handles it)
```

★ KEY TAKEAWAY

100 m is the universal ceiling. T568B is the modern default — both ends matching = straight-through.

Speed & Capacity (1/2)

Three words people confuse — and the difference matters in interviews

BANDWIDTH

How wide the pipe COULD be

The theoretical maximum the medium can carry.

EXAMPLE

1 Gbps NIC sticker

THROUGHPUT

How much ACTUALLY gets through right now

What you measure with iperf or a download speed test.

EXAMPLE

920 Mbps measured

GOODPUT

Of throughput, how much was useful payload

Throughput minus overhead from headers, retransmits, ACKs.

EXAMPLE

880 Mbps useful

★ KEY TAKEAWAY

Marketing sells BANDWIDTH. Users feel THROUGHPUT and LATENCY. Always quote all three.

Speed & Capacity (2/2)

The units and the often-forgotten dimension: latency

UNITS — BIT-PER-SECOND LADDER

bps	kbps	Mbps	Gbps	Tbps
× 1	× 10^3	× 10^6	× 10^9	× 10^{12}

8 bits = 1 byte → a "100 Mbps" link downloads at ~12.5 MB/s

AND DON'T FORGET LATENCY

Bandwidth = pipe WIDTH. Latency = round-trip COMMUTE TIME (what ping measures).

Two "1 Gbps" networks at 5 ms vs 600 ms latency feel COMPLETELY different. Voice/video die first; web pages crawl; bulk downloads barely notice.

→ *Always describe a network with bandwidth + throughput + latency, not just one number.*

★ KEY TAKEAWAY

The pipe-width number lies if you don't pair it with latency. Quote both — every time.

EXERCISE 1 · HOUR 2

GNS3 + VirtualBox Deep Dive

Snapshots, clones, and the eight NIC modes you must know cold

GNS3 — The 5 UI Regions

Where to look when an instructor says "open the toolbox"

1	**Menu Bar**	File · Edit · View · Control · Tools · Help
2	**Devices Toolbox**	Left side. Drag routers, switches, PCs, clouds to the canvas
3	**Workspace**	Center canvas — your topology lives here. Right-click anything to inspect
4	**Topology Summary**	Right side — every node, every link, the green/red status dot
5	**Console / Server / Errors**	Bottom strip — the GNS3 server log; first place you look when a node won't start

★ **KEY TAKEAWAY**

GNS3 is a sandbox where REAL vendor OS images run. The price of the lab is the time you spend learning the UI.

GNS3 — Three Things to Master

Past these three, GNS3 unlocks like a real lab

01

Marketplace

Free appliances

- Cisco IOS images (legal limits apply)
- OPNsense / pfSense firewalls
- Snort / Suricata IDS
- Mininet, Open vSwitch
- Any Linux distro

02

VM Imports

Drop in any VirtualBox / VMware VM as a node

- Treat any VM as a GNS3 device
- Multiple NICs map to GNS3 links
- Great for malware analysis labs
- Great for hardened-target VMs
- Bridges the GUI ↔ guest OS gap

03

Console

Right-click → Console

- Telnet / serial / VNC depending on node
- Run real CLI commands (show ip int br)
- Save configs with copy run start
- Wireshark on a link in 2 clicks
- Test connectivity with ping / traceroute

★ **KEY TAKEAWAY**

Marketplace + VM imports + Console — once these click, GNS3 stops being an obstacle and becomes a multiplier.

VirtualBox — Snapshots

The fastest insurance policy a junior admin can buy themselves

ONLINE

Captures: disk + RAM

Mid-session checkpoint — everything is frozen as-is, including running processes.

Use case: "I'm about to run an unknown installer."

OFFLINE

Captures: disk only

VM is powered off when you take it. Smaller, faster to restore, no RAM state.

Use case: "clean baseline" — fresh install + tools, ready to clone.

★ GOLDEN RULE

Take a snapshot named clean_baseline BEFORE every lab. 30-second roll-back vs 30-minute reinstall.

★ KEY TAKEAWAY

Snapshots turn destructive experiments into safe experiments. A junior who never breaks anything has never learned anything — but only if they can roll back.

VirtualBox — Clones

Two ways to duplicate a VM — and which one to pick

FULL CLONE

Independent copy

Disk is duplicated end-to-end. The clone has zero ties to the parent — delete the parent and the clone keeps working.

Use case: hand the VM to someone else; clone for a long-running production-like role.

LINKED CLONE

Tiny diff-disk pointing back to parent

Only deltas from the parent are stored. 5 linked clones might fit in the disk space of one full clone.

Use case: 5+ identical lab VMs (saves disk and time). Disposable lab boxes.

⚠ FOOTGUN

Always REGENERATE MAC ADDRESSES for all NICs when cloning — otherwise both VMs collide on the network.

★ KEY TAKEAWAY

Clones save DISK and TIME. Use linked clones for class labs; full clones for handoffs.

VirtualBox — NIC Modes (1/2)

Modes 1–4: Internet, lab, bridged, air-gapped

#	Mode	Inet	Host↔VM	VM↔VM	What it does	Use case
1	NAT (default)	✅	❌	❌	VirtualBox runs a tiny private NAT for the VM.	*"Just give my VM Internet."*
2	NAT Network	✅	❌	✅	Like NAT but multiple VMs share the same private subnet.	*Lab + Internet for 2+ VMs.*
3	Bridged	✅	✅	✅	VM gets an IP from your real router via DHCP.	*VM looks like a real LAN device.*
4	Internal Network	❌	❌	✅	Pure VM-to-VM segment named by you. Air-gapped.	*Malware lab — keep traffic OFF the host.*

★ **KEY TAKEAWAY**

Modes 1–4 cover 90% of beginners' needs. Default is NAT — change it on purpose, not by accident.

VirtualBox — NIC Modes (2/2)

Modes 5–8: host-only, generic driver, cloud, not attached

#	Mode	Inet	Host↔VM	VM↔VM	What it does	Use case
5	Host-Only	❌	✅	✅	Host + VMs share a private vbox-only subnet.	*Host SSH into VM, no Internet exposure.*
6	Generic Driver	varies	varies	varies	UDP tunnel between hosts — lets two laptops share a virtual cable.	*Multi-host UDP tunnel (advanced).*
7	Cloud Network	varies	varies	varies	Wraps Oracle Cloud Infrastructure VCN — VM's NIC lives in the cloud.	*OCI integration / hybrid lab.*
8	Not Attached	❌	❌	❌	The NIC exists but has no cable. Simulates yanking the wire.	*Test how a daemon behaves with no network.*

★ **KEY TAKEAWAY**

The NIC mode IS the network for that VM. Pick wrong → either no Internet or your malware reaches the wider world.

LECTURE 2 · HOUR 3

Standards & Reference Models

Why standards exist — OSI, TCP/IP, Hybrid, encapsulation

Why Standards Exist

The 1970s problem and how we got out of it

THE 1970s PROBLEM

Every vendor invented its own networking.

IBM ≠ DEC ≠ Honeywell ≠ Xerox.

Vendor lock-in for life. Pick IBM and you bought IBM forever.

ISO + IEEE + IETF stepped in.

Goal: a SHARED REFERENCE so any vendor's gear can talk to any other vendor's gear.

THE SOLUTION SHAPE

1. Define the JOB at each layer (what L1 does, what L2 does…).
2. Let any vendor build any layer — as long as the INTERFACES match.
3. Publish the spec FREELY (RFCs, ISO/IEC docs, IEEE 802.x).

Result:

A Cisco router + a Juniper switch + a Linux laptop + an iPhone all interoperate.

★ KEY TAKEAWAY

Without standards, the Internet dies. Every byte you send is a layer-cake of agreements among engineers who never met.

The Standards Bodies

Who owns what part of the Internet — by name

Body	Scope	Famous deliverables
IEEE	L1 / L2 (cables, Ethernet, Wi-Fi)	802.3 (Ethernet), 802.11 (Wi-Fi), 802.1Q (VLANs)
IETF	L3+ (the Internet)	TCP, IP, HTTP, DNS — published as RFCs
ISO	Reference models	OSI 7-layer model (ISO/IEC 7498-1)
ITU-T	Telecom carrier infrastructure	G.652 fiber, G.711 voice codec
TIA	Cable categories	TIA-568 — Cat 5e / 6 / 6a / 8 specs
ICANN	DNS / IP allocation	gTLDs, root DNS, IPv4/IPv6 distribution
W3C	The Web	HTML, CSS, ARIA, accessibility

★ **KEY TAKEAWAY**

IEEE owns wires + waves. IETF owns the Internet itself. ISO owns the theory. Knowing who is who scores points in interviews.

RFCs — Where the Internet is Written Down

Numbered specs, free to read, the source of truth for protocols

WHAT IS AN RFC?

Request for Comments — a numbered, public, peer-reviewed spec maintained by the IETF.

WHY YOU CARE

Every protocol you use day-to-day is an RFC. When tools disagree, the RFC wins the argument.

WHERE TO READ

rfc-editor.org → free, searchable, with status (PROPOSED, DRAFT, STANDARD, OBSOLETE).

FAMOUS RFCs

RFC 791	IPv4
RFC 9293	TCP
RFC 1918	Private IPv4 (10.x, 172.16.x, 192.168.x)
RFC 8200	IPv6
RFC 9110	HTTP semantics
RFC 1034/35	DNS — domain names
RFC 5321	SMTP — email
RFC 8446	TLS 1.3

★ KEY TAKEAWAY

RFCs are free homework. Read RFC 1918 once and you'll never confuse 192.168.0.0/16 with anything else.

The OSI Model (1/2)

7 layers — top to bottom — with PDU and example

#	Layer	Job	PDU	Example
7	**Application**	What the user sees	data	HTTP, SSH, DNS
6	**Presentation**	Format, encrypt	data	TLS, JPEG, ASCII
5	**Session**	Open / close conversations	data	RPC, NetBIOS
4	**Transport**	End-to-end + ports	segment	TCP, UDP
3	**Network**	Logical address + route	packet	IP, ICMP, routers
2	**Data Link**	Local address + frame	frame	Ethernet, switches
1	**Physical**	Cables, voltage, light	bits	UTP, fiber, RJ-45

★ **KEY TAKEAWAY**

OSI is the LANGUAGE every networker speaks — even if TCP/IP is the STACK they actually run. Memorize the 7 layers forever.

The OSI Model (2/2)

Two mnemonics that survive interview pressure

BOTTOM → TOP (L1 → L7)

Please Do Not Tell Sales People Anything

P	Physical
D	Data Link
N	Network
T	Transport
S	Session
P	Presentation
A	Application

TOP → BOTTOM (L7 → L1)

All People Seem To Need Data Processing

A	Application
P	Presentation
S	Session
T	Transport
N	Network
D	Data Link
P	Physical

★ KEY TAKEAWAY

Pick ONE mnemonic. Recite it once a day for a week. Then you own the OSI layers for the rest of your career.

TCP/IP & Hybrid (1/2)

TCP/IP — the 4-layer model that actually runs the Internet

TCP/IP (DoD model – 4 layers)

Application	*OSI 5+6+7*	HTTP, DNS, SSH, TLS
Transport	*OSI 4*	TCP, UDP
Internet	*OSI 3*	IP, ICMP
Network Access	*OSI 1+2*	Ethernet, Wi-Fi

WHY ONLY 4 LAYERS?

TCP/IP was DESIGNED before OSI was finished — for the actual ARPANET.

It collapses OSI's L5 + L6 + L7 into a single "Application" layer because real-world apps (HTTP, SSH) handle session, presentation, and application logic together.

It collapses OSI's L1 + L2 into "Network Access" because TCP/IP doesn't care what the wire is.

Result: a leaner, real-world-tested model.

★ KEY TAKEAWAY

TCP/IP is what your laptop actually runs. OSI is how engineers TALK ABOUT what your laptop runs.

TCP/IP & Hybrid (2/2)

Hybrid — the 5-layer teaching sweet spot Cisco NetAcad uses

HYBRID (5 layers)

5	Application	HTTP, DNS, SSH, TLS
4	Transport	TCP, UDP
3	Network	IP, ICMP
2	Data Link	Ethernet, switches
1	Physical	UTP, fiber, voltage

WHY THREE MODELS?

OSI	Theory + troubleshooting language
TCP/IP	What actually runs on the Internet
Hybrid	Best for teaching beginners

Hybrid: keeps L1 / L2 separate (real, testable). Drops Session + Presentation (rarely real layers).

★ **KEY TAKEAWAY**

OSI is THEORY. TCP/IP is PRACTICE. Hybrid is HOW WE TEACH. All three matter — and you'll meet all three at job interviews.

Three Models, Side-by-Side

How OSI, TCP/IP, and Hybrid map to the same physical reality

OSI (7)		TCP/IP (4)		Hybrid (5)	
7	Application	4	Application	5	Application
6	Presentation				
5	Session				
4	Transport	3	Transport	4	Transport
3	Network	2	Internet	3	Network
2	Data Link	1	Network Access	2	Data Link
1	Physical			1	Physical

★ **KEY TAKEAWAY**

Same physical wires. Same protocols. Three different ways of LABELING the same stack — each one optimized for a different audience.

Encapsulation (1/2)

One message, four envelopes — how every byte travels

TRAVELING DOWN THE STACK (sender side)

```
L7  →  "Hello"
L4  →  [TCP hdr] "Hello"                                  (segment)
L3  →  [IP hdr][TCP hdr] "Hello"                           (packet)
L2  →  [Eth hdr][IP hdr][TCP hdr] "Hello" [Eth tail]       (frame)
L1  →  0101110010101000001110100010101010101010101110...   (bits)
```

AT THE RECEIVER

Decapsulation = same in REVERSE. Each layer strips its header and passes the inner payload up.

★ KEY TAKEAWAY

Encapsulation is how all the layers cooperate. Down at the sender, up at the receiver. Same data, different envelope at each step.

Encapsulation (2/2)

The diagnostic ladder — first failed layer wins

WALK UP FROM L1

L1	Cable plugged? Link light? `ip link`
L2	Right VLAN? ARP working? `arp -n / ip neigh`
L3	Right IP? Right gateway? `ip a / ip r / ping gw`
L4	Port open? Firewall? `ss -tlnp / nc -vz`
L7	App responds? DNS resolves? `curl -v / dig`

REAL-LIFE: "THE SITE IS DOWN"

User: "Your site is broken!"

Junior dev: opens the source code at L7…

→ 3 hours wasted debugging templates

Senior NetEng: walks UP from L1 — finds an unplugged switch trunk in 90 seconds.

90% of "the network is down" tickets die at L1 or L2.

★ KEY TAKEAWAY

First failed layer WINS. Always start the troubleshoot at L1. Then L2. Then L3. Don't jump to L7.

EXERCISE 2 · HOUR 4

Project Stage 2 — Build the Topology

Acme SoHo Consulting — 1 router, 1 switch, 2 endpoints, real ping

Project Stage 2 — What You Build

The physical-layer skeleton of Acme SoHo Consulting

GNS3

- 1 router (any image you like)
- 1 multilayer switch
- 2 VPCS endpoints
- All links GREEN — labels visible
- Ping working between the two VPCS

VirtualBox

- 2 linked-clone Ubuntu VMs
- Both NICs on Internal Network 'acme-lan'
- MAC addresses regenerated
- Static IPs in 192.168.50.0/24
- Ping working between the two VMs

Wireshark

- Start capture on the GNS3 link
- Trigger ARP request (ping a fresh IP)
- Capture ARP request + ARP reply
- Capture the resulting ICMP echo + reply
- Save as acme_stage2.pcap

★ **KEY TAKEAWAY**

GNS3 teaches you how a real Cisco router/switch behaves. VirtualBox teaches you how a real OS NIC sees L1 / L2.

Project Stage 2 — Deliverables

What you turn in — written so a stranger could learn from it

01

Annotated screenshots

Green topology in GNS3.
NIC config in VirtualBox.
Successful pings on both stacks.

02

Saved .gns3project portable file

Use File → Export portable project. Keeps the appliance bindings reproducible.

03

.pcap of ARP + ICMP traffic

ARP request + reply, then ICMP echo + reply. Filter: arp or icmp

04

Reflection paragraph

Could a beginner follow my report end-to-end?
If NO — rewrite the section that fails the test.

★ **KEY TAKEAWAY**

You just built Project Stage 2 — the physical-layer skeleton of Acme SoHo. Next session you wire the LAN.

Quick-Reference Recap (1/2)

The 6 things you should never forget from Day 02

01

Cat 5 caps at 100 Mbps

Recable BEFORE you replace anything else.

02

The 100 m rule

Almost every twisted-pair category caps at 100 m.

03

Bandwidth ≠ throughput

And both ≠ MB/s. 8 bits = 1 byte. Always.

04

OSI 7 layers — bottom-up

P D N T S P A. Recite, recite, recite.

05

Encapsulation rule

Down to send, up to receive. First failed layer wins.

06

VirusTotal every installer

Always. Forever. Before you double-click.

★ **KEY TAKEAWAY**

If you remember nothing else from Day 02 — remember these six.

Quick-Reference Recap (2/2)

Your homework before Day 03

01 Read

NetAcad NetEss 3.0
Module 6 (Network Media)
+ Module 5 (Communications Principles)

02 Practice

5 OSI vs TCP/IP mapping exercises
Handwritten — pen + paper.
Paste a photo into your project journal.

03 Build

Finish Exercise 2 at home if you didn't finish in class.
Start the .pcap.
Draft the reflection paragraph.

★ **FINAL TAKEAWAY**

You now have a topology that turns on. Day 3, you give it routing brains.

Coming Up — Day 03

What you'll learn next session

01	**The Network Layer (L3)**	From local Ethernet frames to Internet-scale routing.
02	**IPv4 Addressing**	Dotted decimal, classful vs CIDR, public vs private (RFC 1918).
03	**Subnetting**	How a /24 becomes four /26s — without a calculator.
04	**Routing Tables**	How a router decides which interface to send a packet out of.
05	**Project Stage 3**	Wire the LAN — assign subnets, configure interfaces, real ping across networks.

★ KEY TAKEAWAY

Day 02 = the topology turns on. Day 03 = the topology starts thinking. Read NetEss Module 7 before next class.

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Required reading — IP, copyright, ethics

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Thank You

Day 02 complete · Day 03 next: Network Layer — IP, addresses, routing

From the Industry to the Classroom

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